WASDCAT Games

INDEPENDENT GAME STUDIO

Low Poly Colorizer

Official User Manual

Compact guide for streamlined low-poly asset coloring in Blender and optimized material and texture export for Godot 4.

VERSION	AUTHOR	DATE
1.0.1	Frank Winter	2026-09-12

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Foreword

Low Poly Colorizer (LPC) is a Blender extension designed for fast, intuitive, and resource-efficient color and material styling of 3D models in low-poly aesthetics. Developed for game production at **WASDCAT Games**, it bridges the gap between rapid 3D prototyping in Blender and optimal draw-call performance in modern game engines such as **Godot 4**.

The LPC Principle: References Instead of Material Chaos

Traditional low-poly workflows often struggle with two extremes: either the number of material slots explodes through dozens of individual materials, or static albedo texture atlases prohibit flexible physical properties (PBR).

Low Poly Colorizer solves this through a consistent **reference architecture**:

- **Color Reference (Albedo)**: Each face stores discrete coordinates (X, Y) referencing a dynamically generated palette texture (`lpc_palette.png`).
- **Material Look (PBR Preset)**: Each face references a named preset (*Solid*, *Metallic*, *Emission*, *Clearcoat*), resolved via a compact lookup table (`lpc_preset_lut.png`).
- **A Single Material**: All painted faces share the `lpc_multicolor` material. In the game engine, the painted part of a mesh is **a single surface** – one draw call instead of one per color.
- **Live Propagation**: Adjusting palette saturation or preset roughness later updates **all painted surfaces across the entire project instantly**, without re-traversing meshes.

> **Publication Notice**
> This user manual was typeset and published using the modern Markdown publishing tool markpublish.

1 Installation and Quickstart

1.1 Online Installation via Extensions (Recommended)

Starting with Blender 4.2, Low Poly Colorizer can be installed directly through Blender's extension repository:

1. In Blender, navigate to **Edit** ▸ **Preferences** ▸ **Get Extensions**.
2. Type **Low Poly Colorizer** into the search bar and click **Install**. The extension is automatically downloaded, installed, and registered in the N-Panel.

1.2 Manual Installation (ZIP Archive)

For offline workstations or pre-release builds:

1. In Blender, navigate to **Edit** ▸ **Preferences** ▸ **Get Extensions** (or *Add-ons*).
2. Click the menu icon (▾) in the top right corner and select **Install from Disk…**.
3. Select the downloaded `low_poly_colorizer-x.y.z.zip` archive. The sidebar panel (**N-Panel** ▸ **LPC**) is immediately available.

1.3 Quickstart in 60 Seconds

Once the addon is enabled, you can start painting right away:

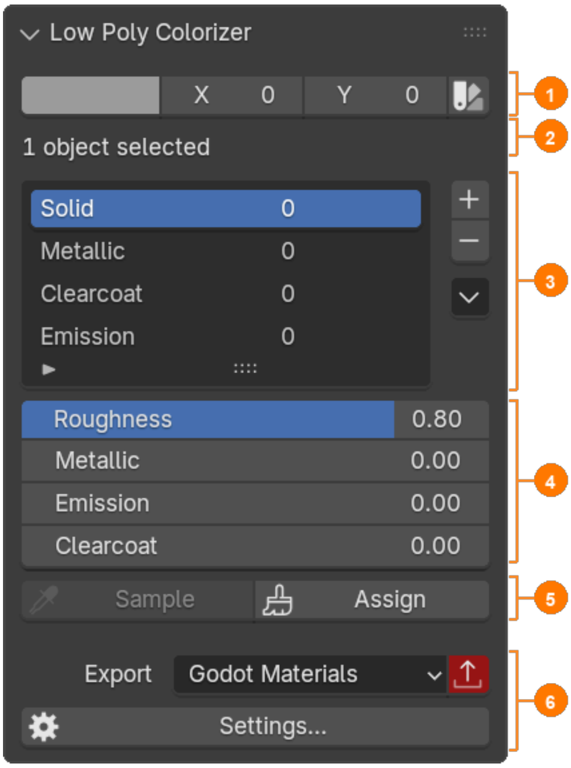
1. **Prepare a model**: Select a mesh object in the 3D Viewport (e.g. the default cube). Optionally press `Tab` to switch to **Edit Mode** and select individual faces.
2. **Open the N-Panel**: Press `N` in the 3D Viewport and switch to the **LPC** tab.
3. **Select a preset**: In the preset list, *Solid* is active by default.
4. **Open the palette**: Click the **Palette button** (color picker) next to the coordinates. The GPU palette overlay opens directly in the viewport.
5. **Assign color**: Click any color cell. The picker closes and the selection is painted in real time with the chosen color and material look.

> **Tip**
> **Live Assignment:** Clicking a cell in the modal picker performs an assignment automatically. Clicking the "Assign" button afterward is not necessary.

2 The User Interface

All controls in Low Poly Colorizer are organized within a compact sidebar panel in the 3D Viewport (**N-Panel ‣ LPC**).

2.1 The N-Panel at a Glance



No.	Section	Description
①	**Color Selection & Picker**	Displays the color and coordinates (X, Y) of the active palette cell. Clicking the **Palette button** (color picker) opens the interactive modal picker in the viewport. Coordinates can also be edited numerically – this paints the current selection immediately, just like a click in the picker.
②	**Selection Status**	Dynamically reports the target selection (*"14 faces in 1 object selected"* or *"1 object se-*

No.	Section	Description
		lected"). Displays a warning if no preset is selected.
③	**Preset List & Actions**	Lists all material presets. The number on the right indicates the **refcount** (how many faces currently use this preset). Includes buttons to add (`+`), delete (`-`), and access the actions menu (`▾`).
④	**PBR Properties**	Controls the physical material properties (*Roughness*, *Metallic*, *Emission*, *Clearcoat*) of the **currently active preset**. Changes propagate instantly to all faces using this preset.
⑤	**Tools**	**Assign**: Assigns the active brush (color + preset) to the selection – selected faces in Edit Mode, whole objects in Object Mode. **Sample**: *(Edit Mode only)* Loads color and preset of the selected faces into the panel. All selected faces must share the same color and preset. **Select / Deselect**: *(Edit Mode only)* Adds all faces with the current color and preset to the selection, or removes them from it.
⑥	**Export & Settings**	Target format selection from modular template sets (default: *Godot Materials*), **Export button** with automatic dirty tracking (turns red when the last export is out of date), and button to open **Settings...**.

2.2 Protection & Feedback Mechanisms

- **Refcount Protection**: A preset cannot be deleted while its reference counter is greater than 0, preventing accidental broken mesh references.
- **Visual Dirty Tracking**: Whenever presets, palette settings, or global values change, the Export button highlights in vibrant **red**. After a successful export, it returns to neutral. Painting faces does not affect it: face data reaches the engine with the model (`.blend`/glTF), not with the export.
- **Color & Preset Matching**: *Select* and *Deselect* use strict logical AND matching: only faces that share **both the exact palette cell and the same preset** are affected.

2.3 Shortcuts & Workflow Hotkeys

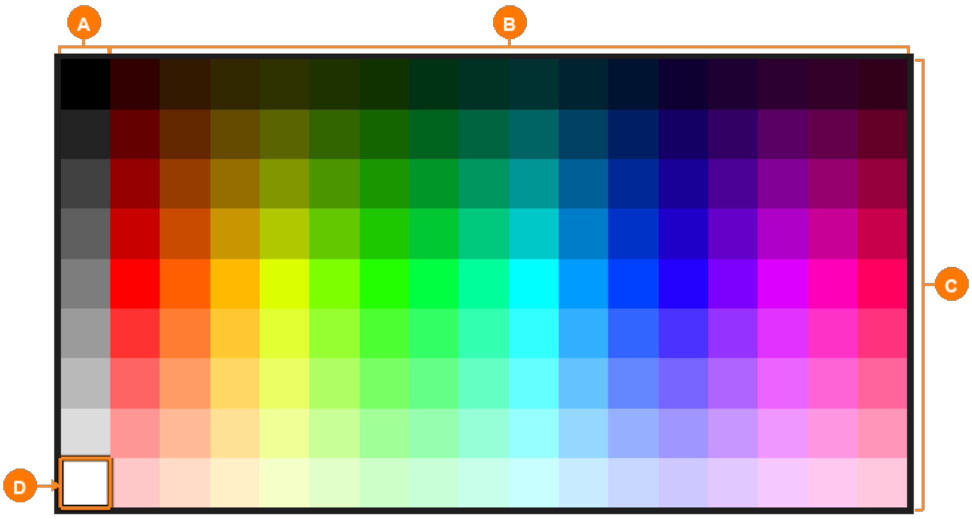
Key	Context	Function
`N`	3D Viewport	Toggle sidebar containing the LPC panel
`Tab`	3D Viewport	Toggle between Object Mode (paint whole object) and Edit Mode
`1` / `2` / `3`	Edit Mode	Switch between Vertex, Edge, and Face selection modes
`L`	Edit Mode (cursor over face)	Select linked geometry island
`A` / `Alt + A`	3D Viewport	Select all / Deselect all

3 Color Palette and PBR Presets

In Low Poly Colorizer, color values (albedo) and surface characteristics (PBR parameters) are managed independently and combined during rendering and shading.

3.1 The Interactive Modal Palette Picker

Clicking the **Palette button** (color picker) in the N-Panel opens the color palette as a hardware-accelerated GPU overlay directly within the 3D Viewport.



Area	Description
Ⓐ	**Greyscale Column (Column 0):** Neutral greys from pure black (top) to pure white (bottom) for monochrome surfaces or neutral shading.
Ⓑ	**Hue Spectrum (Columns 1–16):** Full HSV color wheel from red through yellow, green, cyan, blue, to magenta laid out horizontally.
©	**Brightness & Saturation Gradient (Rows 0–8):** Vertical progression from deep shadow tones (top) through rich saturated colors to pale pastels (bottom). Row 0 is the bottom row.
Ⓓ	**Active Cell & Selection Frame:** Two-tone highlight frame indicating the currently selected cell (X, Y) with instant assignment on click.

The column and row numbers refer to the default grid (greyscale column + 16 hues × 9 rows); both can be changed under *Settings…*.

3.1.1 Navigation within the Picker Overlay

- **Select & Assign**: Click with the **left mouse button (LMB)** on any cell. The coordinates are stored, assigned to the current selection, and the overlay closes.
- **Zoom**: Scroll the **mouse wheel** to zoom continuously into or out of the palette grid (0.25× to 4.0×).
- **Pan**: Hold the **middle mouse button (MMB)** and drag to reposition the palette window anywhere in the viewport.
- **Cancel**: Press **Right-Click**, **Escape**, or click outside the palette to close the overlay without changing the current color selection.

3.2 PBR Presets: Managing Material Looks

Each preset combines four standardized physical material parameters:

- **Roughness (0.0 – 1.0)**: Surface micro-roughness. Lower values yield sharp mirror reflections; higher values produce a soft, diffuse matte look.
- **Metallic (0.0 – 1.0)**: Metallic proportion. 0.0 corresponds to dielectrics (plastic, wood, stone); 1.0 corresponds to true conductive metals.
- **Emission (0.0 – 1.0)**: Surface self-illumination strength, scaled by the global *Emission Factor* (*Settings...*). The cell's albedo color serves as the emission tint.
- **Clearcoat (0.0 – 1.0)**: Secondary protective clear lacquer layer (e.g. automotive paints, varnished surfaces, or wet finishes).

> **Important**
> **Live Update:** Preset sliders do not modify individual mesh vertices or face loops; they update the central lookup table. Adjusting the *Roughness* of the *Solid* preset updates **every face using *Solid* instantaneously**.

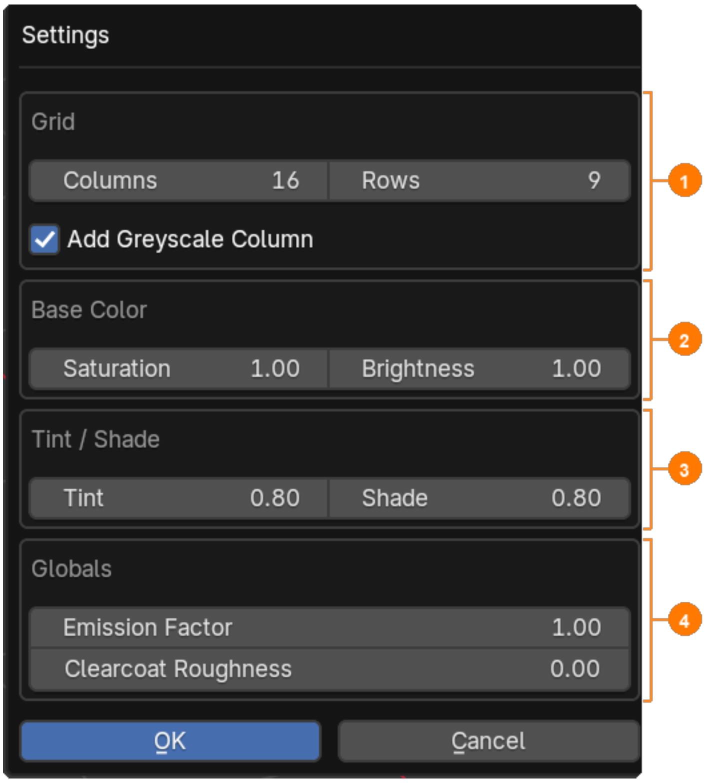
3.2.1 Extended Preset Menu (▾)

- **Duplicate**: Creates a 1:1 duplicate of the active preset to quickly iterate on material variations.
- **Import / Export JSON**: Saves and loads preset definitions as portable JSON files—ideal for sharing material sets across projects. On import, presets with an existing name are updated and new ones are appended.
- **Load Default Presets**: Adds the four bundled factory presets (*Solid*, *Metallic*, *Clearcoat*, *Emission*) if missing and resets existing presets of the same name to the factory values.

- **Create Geometry Node Group**: See chapter *Engine Export and Integration*.

3.3 Global Settings (*Settings...*)

Clicking **Settings...** in the N-Panel opens the configuration dialog for project-wide palette generation and shader parameters. Changes are previewed live; *Cancel* reverts them.



Area	Description
①	**Grid**: Number of hue columns (*Columns*, excluding the greyscale column) and rows (*Rows*), plus a toggle for the monochrome greyscale column (*Add Greyscale Column*).
②	**Base Color**: Base saturation (*Saturation*) and base brightness (*Brightness*) for all hues in the color wheel.
③	**Tint / Shade**: Strength of the vertical gradation steps (highlighting via *Tint*, darkening via *Shade*).
④	**Globals**: Project-wide parameters including the global emission multiplier (*Emission Factor*) and base clearcoat roughness (*Clearcoat Roughness*).

4 Engine Export and Integration

Low Poly Colorizer utilizes a generic, template-based export system (`.tpl`) that is flexibly extensible to other engines. The included default target is **Godot 4** (*Godot Materials*).

4.1 One-Click Export to Godot

LPC exports **no mesh geometry** (models are imported as `.blend` or glTF carrying their `lpc_uv0`/`lpc_uv1` maps). The **Export button** opens a folder dialog and writes prepared materials, shaders, and helper classes into the chosen folder:

- **Textures**: `lpc_palette.png` (albedo/colors) and `lpc_preset_lut.png` (PBR lookup table).
- **Shaders & Materials**: Preconfigured ShaderMaterials (`.tres`), shaders (`.gdshader`), and shared code (`.gdshaderinc`) – in two variants, `lpc_multicolor` and `lpc_singlecolor`.
- **Helper Classes**: GDScript class `lpc_singlecolor_resource.gd` for reusable named looks (palette cell, preset, emission) with a generated `Preset` enum.
- **Import Notes**: `README.md` with the texture import settings Godot requires.
- **Dirty Flag Tracking**: The export button illuminates in red whenever the last export is out of date and resets to neutral once synchronized.

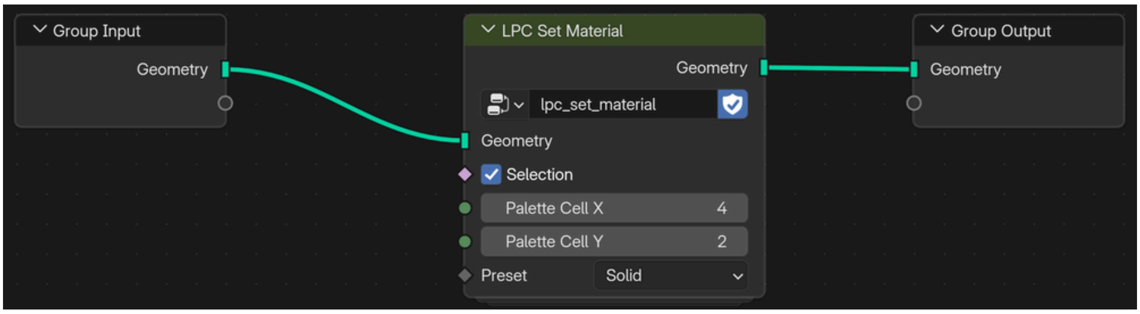
4.2 Using the Files in Godot

1. **Texture import settings**: Both PNGs are data textures, not ordinary color textures. Select each one in the *FileSystem* dock, set the following in the *Import* tab, and click *Reimport*: *Detect 3D* ‣ *Compress To* = **Disabled**, *Compress* ‣ *Mode* = **Lossless**, *Mipmaps* ‣ *Generate* = **Off**. For `lpc_preset_lut.png`, additionally set *Process* ‣ *Fix Alpha Border* = **Off**. The settings survive future re-exports.
2. **Painted meshes**: Assign `lpc_multicolor.tres` as the material override of the imported mesh. It reads color and preset per face from UV/UV2.
3. **Other meshes** (not painted in LPC): Use `lpc_singlecolor.tres`. Palette cell, preset, and emission are set per `MeshInstance3D` as instance shader parameters, so many objects can share this one material and still look different.
4. **Workflow**: Changes to painted faces travel with the model – Godot picks them up when it re-imports the `.blend` or glTF. A new LPC export is only needed when the Export button is red.

4.3 Procedural Workflows with Geometry Nodes

For procedural meshes, generate the `lpc_set_material` node group via the preset menu (▾) by selecting **Create Geometry Node Group**:

1. LPC constructs the node group and attaches it as a Geometry Nodes modifier to the active object.
2. In the Node Editor, drive palette coordinates (X, Y) and material preset procedurally through input sockets. The shared `lpc_multicolor` material handles shading automatically.



4.4 Maintenance: Repairing UV Maps (*Fix UV Maps*)

Low Poly Colorizer utilizes two designated UV channels: `lpc_uv0` (palette color coordinates) and `lpc_uv1` (preset LUT coordinates).

When meshes are joined (`Ctrl + J`), cut with Boolean operations, or imported from external sources, UV layers can become swapped or omitted. In this case, LPC automatically displays a highlighted warning box at the bottom of the N-Panel:

> ⚠ **Warning**
> **UV maps need fixing:** Clicking **Fix UV Maps** (Object Mode only) recreates missing LPC UV maps, moves `lpc_uv0` / `lpc_uv1` to the first two UV slots, and ensures seamless import into Godot. If `lpc_uv0` had to be recreated, the palette colors of the affected faces are lost – LPC reports this, and those faces must be repainted.

4.5 Troubleshooting Checklist

Problem	Cause	Solution
No colors in viewport	3D Viewport is set to *Wireframe* or *Solid*	Press `Z` and switch shading mode to **Mater-**

Problem	Cause	Solution
	mode without texture preview enabled.	**ial Preview** or **Rendered**.
Faces do not color	In Edit Mode, no faces are selected, or no preset is selected in the list.	Select faces with `A` or `L` and ensure a preset is highlighted in the list.
Minus button (-) disabled	The selected preset is currently used by mesh faces (refcount > 0).	Repaint those faces with a different preset. *Select* finds faces with the current color **and** preset – pick the matching color first (e.g. via *Sample*).
Exported look differs in Godot	Textures were not re-exported after modifying presets or palette (red export button).	Click the red **Export button** in the N-Panel to update texture assets.
Colors wrong or presets ignored in Godot	Godot compressed the data textures on import.	Set the texture import settings (see *Using the Files in Godot*) and click *Reimport*.